



Onboarding Databases to DSF Hub Reference Guide

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Amazon RDS for MariaDB Onboarding Steps

This topic reviews the steps required to audit activity on Amazon RDS MariaDB databases and send that information to DSF Hub for auditing, analysis, detecting and preventing security events.

Getting Started

This page contains the information necessary to onboard this data source to Data Security Fabric (DSF) Hub. The following main topics are included:

- An overview of the available scope for the data source, given its native audit data.
- A complete list of prerequisites and permissions that are required for onboarding data sources to DSF Hub.
- Instructions on how to enable audit on the data source and collect data using DSF Hub.
- Initial troubleshooting steps and technical support information.

Onboarding Steps

To ensure a smooth and successful deployment, it is necessary to complete each of these onboarding steps. Please click on each of the steps below to display the content.

✓STEP 1: Onboarding Prerequisites

Onboarding a data source requires preparation, such as gathering permissions and collecting relevant information for your deployment. Assistance may be necessary from a database administrator, network administrator, and an IT administrator to successfully begin monitoring your data source.

Please ensure all of the following items are properly configured or available for use. The information and permissions gathered in this step are required for the remaining onboarding steps.

Cloud Account Prerequisites

Below are the requirements for an Agentless Gateway to communicate with data sources and their related endpoints. Please ensure these are completed by a Cloud Administrator.

Permissions to create or modify AWS resources

Grant the AWS principal the following permissions:

- **rds:CreateOptionGroup** and **rds:ModifyOptionGroup** - To create an Option Group and add 'MARIADB_AUDIT_PLUGIN' option to the option group for enabling audit.
- **rds:ModifyDBInstance** - To attach the Option Group to Amazon RDS for MariaDB instance.

AWS Cloud Account authentication mechanisms

DSF supports the following authentication mechanisms for accessing AWS resources:

- Key pair: for more information on creating a key pair, see [Manage access keys for IAM users](#).
- IAM roles: for more information on creating an IAM role, see [IAM roles for Amazon EC2](#).
- Profile
- Default

The authentication mechanism assigned to the Agentless Gateway should be granted the following privileges:

- To use the DSF Discovery feature:

```
rds:DescribeDBInstances
logs:DescribeLogGroups
```

- To access the CloudWatch Log Group and retrieve audit logs:

```
logs:DescribeLogGroups
logs:DescribeLogStreams
logs:FilterLogEvents
logs:GetLogEvents
```

- To use DSF's Audit Policy Management feature:

```
rds:DescribeDBInstances
rds:DescribeOptionGroups
rds:DescribeDBSecurityGroups
ec2:DescribeSecurityGroups
rds:CopyOptionGroup
rds:ModifyOptionGroup
rds>DeleteOptionGroup
rds:ModifyDBInstance
```

✔STEP 2: Enabling Audit on the Data Source

Once the required permissions and information have been obtained, please complete the following steps to enable audit on the data source.

Auditing on this source can be enabled either via the AWS Portal or AWS CLI. Please choose one of the below methods to enable audit logging.

Enabling Audit via AWS Portal

1. Create an Option Group

Create a new option group using the below steps. To update an existing option group instead, skip to the section "Add 'MARIADB_AUDIT_PLUGIN' to the Option Group".

- On the AWS Portal, go to RDS → Option groups → Create group
- Provide a Name and Description

- Select the "mariadb" Engine from the Engine drop-down list
- Select the same Major Engine Version as your RDS MariaDB instance
- Click on Create

Note:

Create the option group in the same region as the Amazon RDS for MariaDB instance.

2. Attach the Option Group to the MariaDB instance and redirect audit logs to CloudWatch Log Group

For a new MariaDB instance, attach the option group and redirect the audit logs to a CloudWatch Log Group while creating the instance. For more information, see [Creating an Amazon RDS DB instance](#).

For an existing MariaDB instance, follow the below steps to attach the option group to the MariaDB instance and redirect audit logs to a CloudWatch Log Group. For more information, see [Modifying an Amazon RDS DB instance](#).

- Navigate to the Amazon RDS console
- In the navigation pane, select Databases
- Select your Amazon RDS for MariaDB instance
- Select Modify
- Under Additional Configuration:
 - In the Database options → Option group section, select your Option group from the drop-down
 - In the Log exports section, select Audit log and Error log to publish to CloudWatch Logs
- Click on Continue
- Select "Apply Immediately" for the changes to take effect immediately
- Click on Modify DB instance
- The CloudWatch log streams will now start populating in the forms:
 - /aws/rds/instance/<db-instance-name>/audit
 - /aws/rds/instance/<db-instance-name>/error

For more information refer to [Publishing MariaDB logs to Amazon CloudWatch Logs](#).

Enabling Audit via AWS CLI

1. Create an Option Group

Create a new option group using the below command. To update an existing option group instead, skip to the section "Add 'MARIADB_AUDIT_PLUGIN' to the Option Group".

```
aws rds create-option-group \
  --option-group-name <option-group-name> \
  --engine-name <value> \
  --major-engine-version <value> \
  --option-group-description <value>\
```

Note:

Create the option group in the same region as the Amazon RDS for MariaDB instance.

For more configuration information refer to [Create Option Group via CLI](#).

2. Add "MARIADB_AUDIT_PLUGIN" to the Option Group

Edit the option group to add MARIADB_AUDIT_PLUGIN plugin and enable audit logging:

```
aws rds add-option-to-option-group \
  --option-group-name <option-group-name> \
  --options '[{"OptionSettings":[{"Name":"SERVER_AUDIT_EVENTS","Value":"CONNECT, QUERY, TABLE"}]}' \
  --apply-immediately
```

Note:

CONNECT activity (session activity, i.e. logon, logoff, and failed logons) is recorded for all users, regardless of **SERVER_AUDIT_EXCL_USERS**. For more details, see [MariaDB Audit Plugin support](#)

For more configuration information refer to [Add Option to Option Group via CLI](#).

3. Attach the Option Group to the MariaDB instance and redirect audit Logs to CloudWatch Log Group:

```
aws rds modify-db-instance \
  --db-instance-identifier <db-instance-name> \
  --option-group-name <option-group-name> \
  --cloudwatch-logs-export-configuration '{"EnableLogTypes":["error","audit"]}' \
  --apply-immediately
```

For more configuration information refer to [Modify DB Instance via CLI](#).

✓STEP 3: Collecting Audit Data

After completing the prerequisites and enabling audit, the data source is ready to be onboarded onto DSF. This can be accomplished using **ANY ONE** of the methods listed below:

- Importing Assets via Unified Settings Console (USC)
- Importing Assets via an Asset Template Spreadsheet
- Importing Assets via DSF Open APIs

Please use the Asset Specifications documents below as a guide to fill in the field values for this data source and if applicable, its related assets.

[AWS Asset Specifications](#)

[AWS RDS MariaDB Asset Specifications](#)

[AWS Log Group Asset Specifications](#)

Note:

When onboarding an Amazon data source that exports audit data via **Amazon CloudWatch**, three assets are required:

- An **AWS Cloud Account** asset
- A **Database or Instance** asset
- An **AWS Log Group** asset

Define the relationship between these assets using the `parent_asset_id` field (or **Parent Asset Name** when using USC). The asset hierarchy should be as follows:

- The **AWS Cloud Account** is the parent of the **Database or Instance** asset.
- The **Database or Instance** asset is the parent of the **AWS Log Group** asset.

Refer to the asset template spreadsheet for a visual demonstration of the asset structure and relationship. For download instructions, see [Downloading Asset Templates](#).

Importing Assets via Unified Settings Console (USC)

USC allows users to configure a full audit flow, including importing new data assets. To add a new data source asset in USC, please follow these steps:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Data Sources tab to view the data sources of the DSF Hub. Click **Add** to open the Add Data Source form.
3. In the Data Source Type dropdown menu, select a data source.
4. Specific data source configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. If required for the data source, under Connection, select an authentication method (Auth Mechanism) from the dropdown menu. The mandatory fields for the selected Auth Mechanism are displayed; to see optional configuration fields, click Advanced.
6. Under Monitoring, select the Agentless Gateway that will own the data source in the Gateway field, then click **Save**.
7. Locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on adding, viewing and editing assets in USC, see the [Adding Assets via Unified Settings Console \(USC\)](#) documentation.

Discover Assets using Cloud Accounts in USC

The Cloud Accounts tab allows you to run the Discovery playbook at any time to automatically discover and onboard new assets. Follow the instructions below to add a new cloud account and run Discovery via USC.

To add a new cloud account:

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Unified Settings Console**. In the Appliances pane, select **DSF Hub**.
2. Click the Cloud Accounts tab to view the cloud accounts of the DSF Hub. Click **Add** to open the Add Cloud Account form.
3. In the Cloud Account Type dropdown menu, select Amazon, Azure, or Google Cloud.
4. Specific configuration sections will display: Details, Connections, and Monitoring. Configure the mandatory configuration fields under Details and any optional configuration fields displayed under Advanced.
5. Under Connection, select the authentication mechanism that the Agentless Gateway will use to connect to the cloud account.
 - For Amazon: in the Auth Mechanism dropdown menu, select Profile, Key, IAM Role or Default.
 - For Azure: in the Auth Mechanism dropdown menu, select Client Secret, Auth File, or Managed Identity.
 - For Google Cloud: in the Auth Mechanism dropdown menu, select Default or Service Account.
6. Depending on the selected cloud account type and authentication mechanism, different fields will be displayed under the Connection section. Fill them out accordingly.
7. Under Monitoring, select the Agentless Gateway that will own the cloud account in the Gateway field, then click **Save**.

To run Discovery in USC:

1. Locate the desired cloud account asset in USC as described above.
2. Click on the kebab menu icon to the right of the cloud account on which you want to run the Discovery feature.
3. Click **Start Discovery**. Discovery will start running in the background.
4. After Discovery has run successfully, locate the asset you want to connect to the Agentless Gateway. Click on the kebab menu to the right of your asset, then **Enable Audit Collection** to start collecting audit data.

For more information on importing data source assets using cloud accounts in USC, see the [Discovering Data Sources in DSF Hub Cloud Accounts using USC](#) documentation.

Importing Assets via an Asset Template Spreadsheet

Complete these steps to import data source assets using asset template spreadsheets:

1. From the DSF Portal of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specification documentation as a guide for filling in the necessary field values.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **'Validate All'**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.

7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the data source asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect Gateway** to start collecting audit data.

For more information, please see the [Adding Assets via the Import Assets Page](#) documentation.

Note:

Use the following asset templates when importing via spreadsheet:

- Asset template name for Amazon RDS for MariaDB - **AMAZON_RDS_MariaDB_template.xlsx**
- Asset template name for AWS account - choose ONE of the four templates listed below based on the type of authentication mechanism:
 - **AMAZON_EC2_DEFAULT_ROLE_template.xlsx**
 - **AMAZON_IAM_ROLE_template.xlsx**
 - **AMAZON_KEY_PAIR_template.xlsx**
 - **AMAZON_PROFILE_template.xlsx**

Discover Assets using Cloud Accounts in the Assets Dashboard

Note:

This section can be skipped if the data source asset(s) have already been onboarded using spreadsheets in the previous section.

After configuring and onboarding a cloud account asset with the necessary permissions, the Asset Discovery feature can find and onboard related data source assets with just a few clicks.

1. From the DSF homepage of your DSF Hub, under **Apps**, click **Sync Spreadsheet**. In the overlay, click **Import Assets**.
2. On the Import Assets page, go to the **Assets Templates** dropdown menu. Select the template of the cloud account you want to import and click **Download**.
3. Use the Asset Specifications documentation as a guide for completing the asset template spreadsheet for the cloud account asset.
4. On the Import Assets page, go to the section named **Upload 'Assets and Connections to Import' spreadsheet**. Navigate to the spreadsheet that you filled out and click **Open**.
5. To complete the process of onboarding the asset with the asset template spreadsheet, click **Validate All**.
6. Check for any errors or warnings, make any desired changes and then click **Run 'Import Assets'**.
7. Click on the **Assets Dashboard** button to navigate to the Assets Dashboard page, then locate the cloud account asset that was imported.
8. Click the kebab menu to the right of the asset, then click **Discovery**. Choose the data source type to be discovered to start the Discovery playbook.
9. After the Discovery playbook has run successfully, locate the desired data source asset.
10. Click the kebab menu to the right of the asset, then click **Management > Audit Actions > Connect**

Gateway to start collecting audit data.

For more information on discovering assets, please see the [Adding Assets via Asset Discovery](#) documentation.

Importing Assets via DSF Open APIs

Data Security Fabric (DSF) Open APIs provide functions for onboarding and managing assets (log aggregators, cloud accounts, data sources, secret managers and other assets) via a RESTful API. For more information on the supported assets and how to onboard them, please see [Using DSF Open APIs](#).

✓STEP 4: Managing Audit Policies

Audit Policies are created to specify which actions will be collected from the database. This can be done in two ways:

- **Using the Audit Policy Management Feature**
(or)
- **Creating an Audit Policy manually**

Using the Audit Policy Management Feature

DSF provides automated solutions for managing and monitoring audit policies. Audit policies can be managed from the Asset Dashboard (OR) USC.

Managing Audit Policies with Asset Dashboard

To apply an audit policy via Asset Dashboard:

1. In the DSF Portal, under **Dashboards**, click **Asset Dashboard**.
2. In the top left-hand side of the **DSF DataVis** window, click **Dashboard**, and then click the **Asset Details** link. The **Asset Details** dashboard opens.
3. In the **Asset Details** dashboard in DSF DataVis, go to the **Assets** panel and locate the asset to which you want to apply the audit policy.
4. Click the three vertical dots at the end of the row for the asset, in the **Actions** column.
5. In the menu that opens, select **Management > Audit Actions > Turn on Audit Policy**. The **Turn On Audit Policy** window opens.
6. In the **Turn On Audit Policy** window, under **Audit Policy**, select an audit policy by type or by name.
7. Click **Submit**. The audit policy is turned on, that is, applied to the asset.
8. If you want, click **Refresh** to view the asset's audit policy details (e.g. policy_template_name) in the relevant row in the **Assets** panel.

For detailed instructions and supported database versions, see [Managing Audit Policies](#) in the user guide.

Managing Audit Policies with Unified Settings Console

To create and apply a new DSF Hub audit policy via USC:

1. From the main page of **DSF Portal**, under **Apps**, click **Unified Settings Console**.

2. In the **Appliances** pane, select **DSF Hub**.
3. Click the **Audit Policies** tab. The audit policies of the DSF Hub appliance are displayed.
4. Click **Add**.
5. In the **Configuration** tab, in the **Policy Name** field, type a name for the policy.
6. In the **Configuration** tab, select a policy type. Options available in this menu include **Native audit policy**, **Gateway audit policy** and **Combined audit policy**.
7. Select match criteria. Available options depend on audit type selected: For more information about these criteria, see [Supported Criteria for Audit Policies](#).
8. If you selected the **Exclude List** type, add at least one of the following sets:
 - DB User Name
 - Analyzed Client IP
 - Source Program

You can add multiple sets of **Exclude List** criteria via the **+ Add** button.
9. If you selected the **Action Groups** policy type, add one or more action groups. For more information, see [Viewing Lists of Actions for Action Groups](#).
10. Click the **Applied to** tab and select the data source assets to which the policy will be applied.
11. Turn **Audit collection** on or off. This option provides a convenient method to enable audit collection directly from the audit policy.
12. Click **Save**. DSF Hub begins updating the audit policy; this process may take a few minutes to complete.

For detailed instructions, see [Managing Audit Policies via USC](#).

Note:

- The DSF Audit Management feature only supports "ALL" and "SYSTEM" audit policy types for DSF versions 4.10, 4.11 and 4.12 and enables auditing of the following SERVER_AUDIT_EVENTS based on the policy type:
 - **ALL** Policy Type: CONNECT, TABLE, QUERY
 - **SYSTEM** Policy Type: CONNECT, TABLE, QUERY_DDL, QUERY_DCL
- The DSF Audit Management feature also configures and performs the following changes if not already configured:
 - The Option Group setting SERVER_AUDIT_EXCL_USERS is set to rdsadmin
 - MariaDB instance is modified, enabling export of "audit" and "error" audit logs to CloudWatch Log Group

Creating an Audit Policy Manually

Alternatively, it is possible to manually create and manage audit policies on the data source itself.

Add "MARIADB_AUDIT_PLUGIN" to the Option Group

- Select the Option Group and click on Add option
- Select the "MARIADB_AUDIT_PLUGIN" Option name from the drop-down

- Change/verify that the following "Option settings" are assigned the below values:

Option Setting	Value
SERVER_AUDIT_LOGGING	ON
SERVER_AUDIT_EVENTS	CONNECT,QUERY,TABLE
SERVER_AUDIT_EXCL_USERS	rdsadmin

Note:

CONNECT activity (session activity, i.e. logon, logoff, and failed logons) is recorded for all users, regardless of **SERVER_AUDIT_EXCL_USERS**. For more details, see [MariaDB Audit Plugin support](#)

- Click 'Yes' to apply changes immediately
- Click on Add option

For details on configuration options, refer to [Audit Plugin option settings](#).

Troubleshooting

Should you encounter any unexpected issues or behaviors, you may check the status of the following services and associated log files to help pinpoint the root cause. If additional assistance is needed at any time, technical support staff is available to help users of all technical levels via <https://supportportal.thalesgroup.com/csm>.

On the Agentless Gateway host(s) please review the following:

- Gateway log file:

```
$JSONAR_LOGDIR/gateway/cloud/aws/mariadb/sonargateway.log
```

- Run the following command to verify the status of the gateway services and restart if necessary:

```
sudo systemctl status gateway-aws@mariadb.service
```

For more information...

Need Help? For assistance with any DSF Hub or related products, please contact Online Technical Support via <https://supportportal.thalesgroup.com/csm>. A team of technical customer success representatives are ready to assist users of all skill levels.

Related Topics: Below are links with related onboarding information and procedures:

- [Using DSF Open APIs](#): A guide to onboarding and managing assets via DSF Open APIs.
- [Adding assets via Unified Settings Console \(USC\)](#): Instructions to add and edit data source assets via the USC.

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- [Adding Assets via the Import Assets Page](#): Instructions to add data source assets via the Asset Dashboard.
 - [Using the Data Security Fabric Portal](#): An overview of the Data Security Fabric (DSF) portal.
 - [Data Security Coverage Tool](#): Systems and version compatibility with DSF Hub.

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